

Introduction to Python

04 June 2026 17:04

Features of Python

- Python is a high level language. It is a free and open source language.
- It is an interpreted language, as Python programs are executed by an interpreter.
- Python programs are easy to understand as they have a clear defined syntax and relatively simple structures.
- Python is case sensitive. For example Number and number are not same in Python.
- Python is portable and platform independent means it can run on various operating systems and hardware platforms.
- Python has a rich library of predefined functions.
- Python is also helpful in web development. Many popular web services and applications are built using Python.
- Python uses indentation for blocks and nested blocks.

It was initially designed by Guido van Rossum in 1991 and developed by the Python Software Foundation. The language was conceived in the late 1980s, and its implementation began in December 1989 at Centrum Wiskunde & Informatica (CWI) in the Netherlands.

Working with Python

To write and run (execute) a program. We need to have a Python interpreter installed in our computer. Or we can use any online Python Interpreter.

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> 3 + 8
11
>>> print(4 + 5)
9
>>> print('Hello!')
Hello!
>>>
```

REPL → Loop → wait for the next command until the session ends.

Read → Accepts user input and parses it into Python expression.

Evaluate → Execute the parse code

Print → Displays the result of execution.

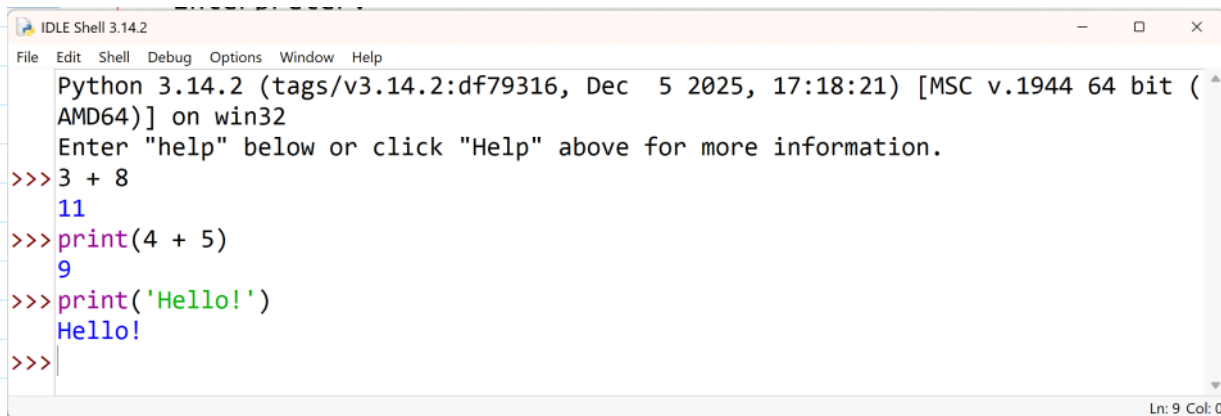
REPL tool.

Python interpreter or shell.

Execution Mode: → Interactive mode.
→ Script mode.

- **Interactive Mode:** To work in the interactive mode, we can simply type a Python statement on the > > > Prompt directly. As soon as we press enter, the

interpreter executes the statement and displays the results.



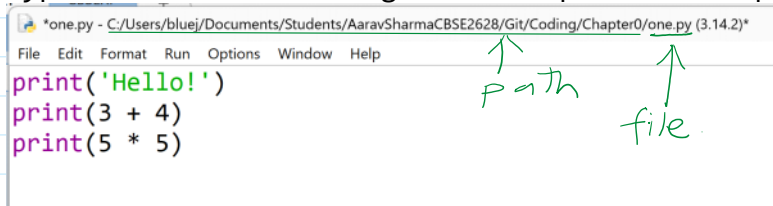
```
IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> 3 + 8
11
>>> print(4 + 5)
9
>>> print('Hello!')
Hello!
>>>
```

Working in the interactive mode is convenient for testing a single line code for instant execution. But in the interactive mode, we cannot save the statements for future use and we have to retype the statements to run them again.

• Script Mode

In the script mode, we can write a Python program in a file, save it and then use the interpreter to execute it. Python scripts are saved as files where filename has extension ".py". By default, the Python scripts are saved in the Python installation folder. To execute a script we can either:

- Type the file name along with the path at the prompt.



```
*one.py - C:/Users/bluej/Documents/Students/AaravSharmaCBSE2628/Git/Coding/Chapter0/one.py (3.14.2)*
File Edit Format Run Options Window Help
print('Hello!')
print(3 + 4)
print(5 * 5)
```

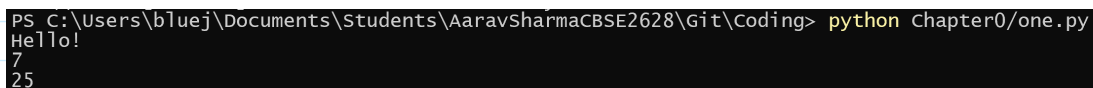
↑ Path ↑ file

Source code

Output:

```
>>>
= RESTART: C:/Users/bluej/Documents/Students/AaravSharmaCBSE2628/Git/Coding/Chapter0/one.py
Hello!
7
25
>>>
```

Or



```
PS C:\Users\bluej\Documents\Students\AaravSharmaCBSE2628\Git\Coding> python Chapter0/one.py
Hello!
7
25
```

PYTHON Keywords

Keywords are reserved words. Each keyword has a specific meaning to Python interpreter and we can use a keyword in our program only for the purpose for which it has been defined. Tika as Python is case sensitive, keywords must be written exactly as given.

import keyword

```
print("Printing All Python keywords")
print("Python keywords are: ")
print(keyword.kwlist)
```

Output:

```
Printing All Python keywords
Python keywords are:
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break',
'class', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for',
'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not',
'or', 'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
```

```
import keyword
```

```
print("Printing All Python keywords")
print("Python keywords are: ")
keyWordlist = keyword.kwlist
print(keyWordlist)
print("Number of keyword:",len(keyWordlist))
```

= RESTART:

```
C:/Users/bluej/Documents/Students/AaravSharmaCBSE2628/Git/Coding/Chapter0/keywo
rd.py
```

```
Printing All Python keywords
```

```
Python keywords are:
```

```
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break',
'class', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for',
'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not',
'or', 'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
```

```
Number of keyword: 35
```